

VMAF-Aware Runtime Shielding for Constrained Deep Reinforcement Learning in Adaptive Video Streaming: A Strictly Pareto-Improving Projection

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Abstract

Client-side Adaptive Bitrate (ABR) control must balance perceptual quality against playback-stall risk, yet purely reward-maximizing deep reinforcement learning (DRL) often learns aggressive bitrate choices that inflate tail stall risk under throughput volatility, while always-on safety projections layered on top of such policies trade away visual quality in opaque, non-tunable ways. This paper revisits DASH/HLS bitrate selection under a constrained Markov Decision Process (CMDP) viewpoint: maximize expected visual quality subject to empirical budgets on rebuffer accumulation and quality switching (*smoothness*), using a primal-dual Lagrangian relaxation trained with Proximal Policy Optimization (PPO). The training stack consumes informative client-side lookahead (near-future chunk information) and Lyapunov-style buffer-regularization cues that stabilize exploration in trace-driven rollouts. On top of the learned policy we contribute a *VMAF-aware Pareto-tunable runtime shield*: a deterministic action projection that, when the catastrophic-download test fires, selects the highest-VMAF rung satisfying a buffer-margin viability constraint subject to a small VMAF-loss budget rather than uniformly decrementing the bitrate index, and whose deployment-time *catastrophic ratio* knob traces an entire operating front without retraining. Our headline empirical finding is a *strict three-way Pareto improvement*: at the recommended operating point (VMAF-loss tolerance

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$= 1.0$), the proposed shield simultaneously increases mean session QoE by +42.13, reduces mean rebuffering by -0.55 %, and increases mean VMAF by +0.22, all paired against the standard index-decrement legacy shield on the *same* policy and trace seeds (paired Wilcoxon, $n = 80$; $p = 5.0 \times 10^{-3}$ for QoE, $p = 2.9 \times 10^{-3}$ for rebuffer). Tightening the VMAF-loss budget further to tolerance = 0.8 widens the win to +64.82 paired QoE, -0.69 % rebuffering, and + + 0.32 VMAF, all simultaneously and statistically significant. A controlled sweep over 47 shield variants—spanning VMAF-aware projection, H -step lookahead rollouts, risk-gating, and catastrophic-ratio sweeps—collapses into only 11 behavioral classes, surfacing the VMAF-aware fallback (rather than the threshold knob alone) as the dominant Pareto driver. Removing the shield from a co-designed policy at inference inflates rebuffering by $|-5.92|$ % ($p = 1.4 \times 10^{-10}$), confirming that the shield is operationally indispensable and that policy/shield co-design is the operationally faithful pattern. Against widely cited baselines under identical evaluation traces, the same operating point reduces mean rebuffering by 10.00 % versus Pensieve and 25.71 % versus RobustMPC. The contribution is therefore a deployable, auditable, and *strictly Pareto-improving* runtime shield with a single tunable knob.

Keywords: Adaptive Bitrate Streaming, Deep Reinforcement Learning, Constrained MDP, Lagrangian Duality, Runtime Shielding, VMAF-Aware Projection, Pareto-Improving Safety, Policy-Shield Co-Design, Video QoE

1. Introduction

HTTP-based adaptive streaming is the default delivery mode for Internet video: clients fetch time-indexed segments from ordinary web caches and CDNs, while a local *adaptive bitrate* (ABR) controller picks a representation (bitrate/resolution) for each short playback interval under time-varying throughput [1, 2]. Because stall events are acutely salient to viewers and quality swings are distracting, modern systems must optimize a *vector* of competing objectives—throughput efficiency, visual fidelity, rebuffering, and stability—rather than myopically maximizing a single scalar signal [3]. Industry-scale demand for mobile and fixed broadband video further raises the stakes for robust client-side control when links are volatile or capacity-constrained [4].

Operationally, the ABR policy observes partial state (buffer occupancy,

recent throughput, manifest metadata) and issues a discrete action each *chunk* period; download time couples stochastically to unknown future throughput, so the plant is inherently non-stationary. Surveys consistently cluster solutions into buffer/threshold heuristics (e.g., BBA [5]), model-predictive style controllers (RobustMPC [6]), and learning-based policies (Pensieve [7] and follow-ups) that replace hand-tuned rules with function approximation [2, 3]. Perceptual reporting increasingly relies on objective scores such as VMAF [8], while evaluation still commonly linearizes trade-offs into a QoE surrogate that penalizes rebuffering and quality variation—a reporting convention we align with for comparability to widely cited MPC/DRL baselines.

Despite strong average-case performance, purely reward-maximizing deep RL can overfit simulators and learn behavior that is aggressive in the *tail*: brief stalls under stress or sharp oscillations that are under-weighted in expectation but harmful in practice [7, 9]. Classical MPC mitigates some pathologies by explicit lookahead, yet its effectiveness depends on throughput predictors that may fail under distribution shift or multi-second volatility [6, 10]. From a *safe decision-making* perspective, this gap naturally suggests *constrained* objectives: maintain high expected quality while keeping distributional proxies of stall and smoothness below operator-chosen budgets—a storyline aligned with constrained Markov Decision Processes (CMDPs) [11] and their Lagrangian / primal–dual surrogates studied in RL theory [12]. A complementary deployment-time guard is the *runtime shield* [13]: a deterministic projection that maps unsafe actions onto a feasible subset before execution. Existing ABR shields, however, hard-wire a *single* risk threshold and a single bitrate-decrement policy, presenting operators with a take-it-or-leave-it operating point and trading away QoE in opaque, non-tunable ways. This paper argues that the right deployment artifact is not a single shield but a *Pareto-tunable shield family*: a shield with one interpretable knob (the *catastrophic ratio*) that traverses the QoE-vs-stall trade space at inference time, complemented by a per-video VMAF-aware fallback so that, when projection is necessary, the shield prefers steps that lose less perceptual quality.

Technical positioning. We *do not* claim formal worst-case safety certificates for neural policies; instead we pursue an engineering-realistic combination that is reproducible under a standardized simulator harness: **(i) Lagrangian CMDP-style training**, where dual variables adaptively reweight stall and smoothing penalties estimated from rollout statistics (projected ascent), instantiated with Proximal Policy Optimization (PPO) [14] and closely related constrained policy learning ideas [15]; **(ii) future-aware observa-**

tions exploiting manifest-available near-future segment features (sizes, complexity indicators such as SI/TI) [1]; **(iii) Pareto-tunable runtime shielding**, where a single deployment-time scalar—the *catastrophic ratio*—rather than a fixed projection rule traverses the QoE-vs-stall front, augmented by a per-video VMAF-aware fallback so that emergency step-downs prefer the highest-quality safe rung rather than uniform decrement [8, 13]; and **(iv) a structural characterization of the shield-design space**: across 47 shield variants spanning VMAF-aware projection, lookahead rollouts, risk-gating, and threshold sweeps, only 11 behavioral classes emerge in our evaluation, isolating the *shape* of the projection rule (whether the VMAF-aware fallback is enabled) rather than the *level* of the catastrophic-ratio threshold or auxiliary lookahead heuristics as the dominant Pareto driver. This split mirrors a practical deployment pattern: approximate constraint satisfaction during training, then add an analyzable, *retunable* inference-time guard whose intervention rate and operating point are both measurable. Accordingly, we evaluate the method as a tunable *stall-risk reduction* mechanism with measured QoE cost, not as a claim of universal scalar-QoE superiority over every ABR baseline.

Lyapunov-style cues (scope). To temper exploration when buffers are dangerously low or throughput estimates are pessimistic, we optionally consume simulator-exposed auxiliary signals resembling drift-aware regularizers [16]. We treat these signals as heuristic stabilizers (not substitutes for rigorous queue stability guarantees under neural approximation), which we state explicitly because reviewers often probe the boundary between motivational queueing intuition and RL claims [16].

Contributions. Relative to MPC baselines without learned predictors [6], classical DRL ABR [7, 17], and QoE-centric safe RL studies [18, 19], this paper contributes:

1. **An end-to-end CMDP+Lagrangian instantiation for adaptive streaming**: session control is cast as maximizing expected fidelity subject to smoothed surrogates of mean stall and bitrate switching, solved via primal–dual-style dual updates coherent with constrained RL duality [12] on top of a future-aware Pensieve-class state encoder.
2. **A VMAF-aware runtime shield with strict three-way Pareto dominance over the standard projection**: when the catastrophic-download test fires, the shield selects the highest-VMAF rung satisfying a buffer-margin viability constraint subject to a small VMAF-loss

budget, rather than uniformly decrementing the bitrate index. On the same trained policy and identical trace seeds, this raises mean QoE by +42.13, lowers mean rebuffering by $|-0.55|$ %, and raises mean VMAF by +0.22 *simultaneously* and statistically significantly (paired Wilcoxon, $n = 80$). To our knowledge no prior ABR shield in the literature reports a strict three-way Pareto improvement of this form [8, 13, 18].

3. **A deployment-time tunable family with two interpretable knobs:** a VMAF-loss tolerance τ and the catastrophic ratio ρ_{cat} . Tightening τ from 1.0 to 0.8 widens the paired win to +64.82 QoE / -0.69 % rebuffer / $++0.32$ VMAF, while ρ_{cat} traces an additional axis of the trade space, all without retraining.
4. **A structural finding about shield-design space:** a controlled sweep over 47 shield configurations (VMAF-aware projection, H -step lookahead rollouts, risk-gating, and threshold variation) collapses into only 11 behavioral classes. We empirically isolate the VMAF-aware fallback (the projection *shape*), not the catastrophic-ratio threshold or lookahead horizon (the threshold *level* or pre-gating heuristics), as the dominant Pareto driver—a finding that simplifies shield engineering practice in production deployments.
5. **Empirical evidence for policy/shield co-design as a deployment pattern:** removing the shield from a co-designed policy at inference inflates paired rebuffering by $|-5.92|$ % ($p = 1.4 \times 10^{-10}$), so the policy and its shield must be shipped together. We argue this is operationally faithful and contrast it with the prevailing practice of bolt-on shielding of unconstrained policies.

Road map. Section 2 situates heuristic, MPC, and learning-based ABR and connects to constrained/safe RL; Section 3 formalizes QoE surrogates and CMDP constraints; Section 4 details Lagrangian PPO training, shields, and algorithmic workflow; Section 5 answers RQs with aggregate tables, statistical tests, and figures; Sections 6–7 discuss threats and summarize.

2. Related Work

We organize prior art along axes that mirror our design choices: classical QoE-centric control, MPC-style lookahead, purely learned bitrate policies, constrained/safe RL, and orthogonal system-level streaming research

(edge-assisted delivery, QoE proxies). Table 1 then condenses representative paradigms and positions our instantiation.

2.1. HTTP adaptive streaming & QoE-oriented control

MPEG-DASH and similar standards decouple packaging from transport: the client obtains a ladder of representations plus chunk timing and nominally runs a bitrate policy purely from manifest-accessible cues and observable downloads [1]. Surveys document a large design space spanning buffer rules, predictor-assisted MPC, and learning [2, 3]. QoE surrogates typically trade average perceptual fidelity (bitrate or VMAF [8]) against stalls and instability; this multi-objective viewpoint is foundational for both classical and learning-based controllers [3].

Heuristic & optimization-flavored controllers. Buffer-based policies such as BBA [5] demonstrate that simple occupancy rules can be surprisingly effective at scale, but they do not explicitly encode fine-grained horizon optimization. BOLA reformulates ABR as an online optimization problem with provable structure under a Lyapunov-style stability argument for buffer evolution [20]: this is spiritually related to drift-based safe RL [21] but instantiated with model-specific assumptions distinct from neural CMDP+PPO wrappers. **Throughput prediction & parameter adaptation.** Throughput predictors underpin many MPC and hybrid designs; CS2P [10] highlights data-driven forecasting as first-class coupling to bitrate selection quality. **Systems-level orthogonal threads.** Systems work on edge-assisted delivery [22–24] and parameter auto-tuning (Oboe [25]) addresses *where* bitrate decisions execute or *how* knobs are calibrated, complementary to declaring explicit stall/smooth budgets on a neural policy itself.

2.2. Model-predictive ABR & lookahead rationale

RobustMPC [6] embodies the dominant MPC narrative in ABR: solve a constrained short-horizon program under an estimated throughput trajectory. When predictors track reality, MPC yields interpretable trade-offs; when they do not, conservative or aggressive mis-selection can appear—especially under bursty mobile links [10, 26]. **Hybrid control–learning.** Recent hybrid pipelines combine differentiable or structured predictors with learning [27]; our framing differs by centering primal–dual constraint handling on rollout statistics [12] while retaining manifest lookahead features similar in spirit (though not identical in mechanism) to MPC horizon inputs [6].

2.3. Deep reinforcement learning for ABR

Pensieve [7] popularized RL on trace-driven emulation with recurrent throughput history and QoE-motivated rewards, catalyzing a family of successors (e.g., D-DASH contemporaneous deep RL formulations [17]). **Strengths.** RL naturally consumes high-dimensional histories and heterogeneous manifest metadata without brittle hand-engineered surrogates for every impairment [9]. **Tensions for multimedia reviewers.** Pure scalar rewards make stall and smoothness *implicit*; multi-objective trade-offs then depend on reward scaling and simulator misspecification. Field evidence (Fugu / “Learning in Situ”) stresses deployment gaps between offline traces and live networks [28], motivating transparent reporting alongside average QoE. These observations motivate declarative surrogates for stall/smooth budgets at training time [18]—distinct from swapping in a heavier reward weight by hand [2].

2.4. Constrained, risk-aware, $\mathcal{E}$ shielded RL

CMDPs formulate expectations with auxiliary budget constraints [11]; primal–dual and Lagrangian relaxations underpin practical RL with constraint feedback [12]. Policy-search analogues tighten trust regions under constraint surrogates [15]. **Safety narratives.** Tutorial-style taxonomies categorize constraint satisfaction, pessimism, shields, Lyapunov methods, and more [19]. Lyapunov-style penalties appear as principled scaffolding in some safe RL recipes [21]; multimedia-specific treatments increasingly adopt CMDP viewpoints for QoE-sensitive streaming [18].

Shields. Logical/runtime shielding prunes or repairs actions before execution to respect specifications [13]. For streaming, shields can be viewed as lightweight *inference-time* stabilizers whose intervention frequency is measurable—a deployment-friendly complement to internal reward shaping when tail stall must be reduced without discarding the learned policy wholesale.

2.5. Positioning of this work

Relative to classical MPC [6], we replace hand-rolled horizon optimization with a learned policy but retain explicit constraint signaling through Lagrangian duals [12] and manifest lookahead comparable to client-visible metadata [1]. Relative to Pensieve-style DRL [7, 17], we externalize stall/smoothness *budgets* rather than burying them inside a single tuned scalar return. Relative to QoE-aware safe RL works in streaming [18], we emphasize a reproducible split: dual updates during training plus deterministic shields at inference with logged intervention rates [13], aligning evaluation with both QoE and

stall/switch summaries [3]. We deliberately do *not* blur these contributions with orthogonal systems optimizations (edge assistance [23, 24], knob auto-tuning [25]) although such layers could stack atop our controllers in future deployments.

Table 1: Qualitative taxonomy of representative client-side ABR paradigms (symbol legend below the grid).

Family (ref.)	Summarized objective / setting	Stall/ smooth budget	Manifest lookahead	Neural bitrate adapt	Runtime determ. guard
BBA [5]	Thresholds on buffer occupancy	Ind.	Imp. sizes	N	Heuristic thresh- olds
BOLA [20]	Online Lyapunov-structured bitrate utility	Imp. stabil- ity lens	Implicit (buffer cou- pling)	N	N
RobustMPC [6]	Short-horizon MPC under predicted throughput	Ind.	Horizon chunk sizes	N	N
Oboe [25]	System-level ABR parameter auto-tuning	N	N	N	N
Pensieve [7]	Scalar RL reward (QoE surrogate)	NF	Var. (blind- ing op- tional)	Y	N
D-DASH [17]	DRL on DASH traces	NF	Manifest ladder fea- tures	Y	N
Fugu / in-situ [28]	Field RL on live sessions	Imp.	Standard client ob- serv- ables	Y	N
QoE-aware safe RL [18]	Constrained / safe RL for streaming QoE	Focal	State design depend- ent	Y	Ctx
Hybrid control- learning [27]	Structured predictors plus learning	Var.	Often predictor- driven	P	N
Ours (Lag. PPO + optional shield)	Primal-dual CMDP surrogate trained with PPO [12, 14]	Adapt. dual λ	MLP policy + mani- fest LA [1, 7]	Y	Shield (opt.) [13]

Column symbols. **Y**=feature clearly present *in spirit*; **N**=not focal; **Ind.**=indirect via rewards/structure; **Imp.**=implicit through dynamics/buffer coupling; **NF**=no declarative constrained stall/smooth budget like ours; **Var.**=configuration-dependent; **P**=partial hybrid coupling; **Ctx**=context-dependent guarding. BOLA: analytic Lyapunov stabilization [20] (not neural CMDP).

3. System Model and Problem Formulation

3.1. Client-Side Abstraction

We adopt the standard chunked HTTP adaptive-streaming abstraction [1, 6, 7]: representations on a bitrate ladder $\mathcal{B} = \{1, \dots, L\}$ index chunk sizes signaled in the manifest; the client observes partial state s_k at chunk boundary k , commits an action $a_k \in \mathcal{B}$, then downloads encoded media for horizon chunk $k+1$ under an unknown time-varying channel. Decisions evolve on the *chunk* clock (typical playback duration T_c , e.g., $T_c=4$ s in our emulation), aligning controller design with MPC/DRL benchmarks in the Pensieve lineage [6, 7].

3.2. Chunk-Timescale Plant Dynamics

3.2.1. Throughput and Latency

Let $\hat{C}_k > 0$ denote the effective empirical throughput allocated to chunk $k+1$ downloads (matching the estimator used by emulation/baselines to keep comparisons apples-to-apples). Choosing representation a_k induces a chunk payload size $\sigma_k(a_k)$ inferred from ladder metadata; with throughput clamped $\tilde{C}_k = \max\{\hat{C}_k, \epsilon\}$ for a tiny $\epsilon > 0$ to avoid division blow-ups [6], nominal download seconds are $d_k = \sigma_k(a_k)/\tilde{C}_k$.

3.2.2. Buffer and Stall

Let $B_k \geq 0$ be buffer occupancy at the start of download k , measured in *seconds of media consumed at nominal rate* [6, 7]. While fetching chunk $k+1$, stall seconds accumulate when playback cannot consume media faster than arrivals:

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}. \quad (1)$$

After the k^{th} interaction completes under this abstracted fluid model [6], nominal buffer evolves as

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c, \quad (2)$$

i.e., buffer drains by download time (stall captured above) and refills by one chunk of playback credit when the next segment is ready. We intentionally keep B_k in seconds because it aligns with QoE surrogates (stall vs. bitrate trade-offs [2]) and MPC-style constraints [6].

3.2.3. Observation–Action MDP Skeleton

Transitions are stochastic through $\hat{C}_k$ sampled from curated traces [2, 26, 29]; deterministic given $(s_k, a_k, \hat{C}_k)$ under σ_k and buffering rules (2). Formal CMDP formulations treat constraint costs as additional signals on transitions [11]; our implementation aggregates them per episode for dual updates [12].

3.3. Evaluation QoE Surrogate

Separating benchmarking from optimization. Controlled experiments still report a *scalar per-chunk QoE surrogate* analogous to RobustMPC/Pensieve disclosures [6, 7]:

$$\text{QoE}_k^{(\text{rep})} = q_k - \alpha \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (3)$$

where q_k denotes instantaneous fidelity (VMAF here [8]), $\text{smooth}_k = |q_k - q_{k-1}|$ proxies quality switching nuisance, α, β match simulator coefficients used by baselines, and smooth_k adopts boundary conventions consistent with their shared evaluation harness. Tables and figures aggregate $\sum_{k \in \text{ep}} \text{QoE}_k^{(\text{rep})}$ (“session QoE”) to align with community practice while our *optimization* objective below keeps q_k in the primal term and separates stall/smooth surrogates into constraints/costs [12].

3.4. Constrained Stochastic Control

We aim to maximize fidelity along trajectories induced by π_θ while explicitly bounding expected stall/smooth surrogates per session:

$$\max_{\pi_\theta} \mathbb{E}_{\pi_\theta, \rho} \left[\sum_{k \in \mathcal{K}_e} q_k \right], \quad (4)$$

$$\text{s.t. } \mathbb{E}_{\pi_\theta, \rho} [\text{RB}(e)] \leq \delta_{\text{rf}}, \quad (5)$$

$$\mathbb{E}_{\pi_\theta, \rho} [\text{SM}(e)] \leq \delta_{\text{sm}}, \quad (6)$$

indexed over stochastic episodes e drawn i.i.d. from ρ (videos $\times$ trace partitions). $\mathcal{K}_e = \{1, \dots, K_e\}$ indexes chunk spans per episode. **Normalized rebuffer.** Consistent with our reporting tables, let total playback-time $T_e^{\text{play}} = K_e T_c$. Define

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} \tau_k^{\text{stall}}, \quad (7)$$

i.e., *stall fraction of session wall-clock*; scaling $\text{RB} \times 100$ matches headline “mean rebuffer ratio (%)” in Section 5.

Smoothness surrogate. We adopt the fidelity-jump analogue used during training/logging:

$$\text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} \text{smooth}_k. \quad (8)$$

Targets $\delta_{\text{rf}}, \delta_{\text{sm}}$ operationalize stakeholder preferences (“keep mean stall below $x\%$ while limiting oscillation”); we treat them as fixed hyper-parameters for reproducibility, mirroring MPC-style QoE knobs [6].

Relaxation caveat. Equation (4)–(6) summarizes the stochastic program motivating our Lagrangian PPO instantiation; empirical CMDP+PPO minimizes sample estimates with warm-started duals and clipped trust regions [14, 15], so finite-sample residuals can remain until additional regularization (Sec. 4) activates.

3.5. Lagrangian Relaxation and Dual Ascent

Penalty form. Introducing duals $\lambda_r, \lambda_s \geq 0$, surrogates $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ denote per-step penalty channels derived from instantaneous stall τ_k^{stall} and smoothness smooth_k so rolling them up coincides episode-wise with $\text{RB}(e)$ and $\text{SM}(e)$ up to fixed normalization factors shared across baselines in the codebase:

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta, \rho}} \left[\sum_{k \in \mathcal{K}_e} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k) \right], \quad (9)$$

with $\widetilde{\text{RB}}_k$ identified with τ_k^{stall} normalization consistent with rollout logging and $\widetilde{\text{SM}}_k$ mirroring smooth_k up to clipping/scaling conventions shared with baselines. Each training episode terminates with empirical constraint estimates $\hat{\text{RB}}, \hat{\text{SM}}$; deficits

$$g_r = \hat{\text{RB}} - \delta_{\text{rf}}, \quad g_s = \hat{\text{SM}} - \delta_{\text{sm}}, \quad (10)$$

drive projected stochastic dual ascent [11, 12],

$$\lambda_{\bullet} \leftarrow \Pi_{\Lambda} [\lambda_{\bullet} + \eta_{\bullet} g_{\bullet}], \quad \Lambda = [0, \lambda_r^{\max}] \times [0, \lambda_s^{\max}], \quad (11)$$

with possibly asymmetric (η_r, η_s) to prioritize stall satisfaction (mirroring operational preference for avoiding stalls over aggressive quality swings).

Remark (auxiliary Lyapunov cues). Training optionally subtracts buffer-deviation and simulator Lyapunov penalty channels [16] as *non-constraint* stabilizers; they do not replace certificates for neural policies and are studied alongside ablations in Section 5.

Remark (runtime shields). Shields $\mathcal{S}$ from Section 4 act as deterministic maps $(s_k, a_k^{\text{raw}}) \mapsto a_k^{\text{exec}}$ before environment stepping; they are *not* additional transition noise but explicit safety-style guards whose intervention frequency is logged for interpretability [13].

4. Method: Lagrangian PPO and Runtime Shields

We realize the CMDP relaxation of Sec. 3 with a *primal-dual* loop compatible with large-scale policy optimization [12, 15]: PPO [14] acts as the primal optimizer on a shaped Markovian reward, while episodic statistics drive projected dual ascent on (λ_r, λ_s) . Runtime shields implement a separate, deterministic action map layered *after* (or, optionally, around) the policy during train/eval, following the “train then guard” narrative common in shielded RL [13, 19]. All quantities share the chunk clock and symbols of Sec. 3: $k = 1, \dots, K_e$, T_c , $a_k \in \mathcal{B}$, B_k , τ_k^{stall} , buffering (2), and surrogates $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$ tied to simulator logs (VMAF-based q_k , instantaneous rebuffer penalties, VMAF-switch penalties).

4.1. Observation and Policy Interface

Observation. Vector s_k concatenates three semantically isolated blocks analogous to Pensieve [7] yet augmented with lookahead metadata advocated for DASH clients [1]: (i) *past throughput* represented as a bounded normalized history; (ii) *control state*: buffer fullness, bitrate index a_{k-1} , and pacing cues exposed by our emulator wrapper; (iii) *near-future manifest features*: bitrate ladder chunk sizes plus spatial/temporal complexity tags (SI/TI) aligned with MPC-style horizon inputs [6]. This stack remains realizable purely from manifests and client-visible measurements [1].

Action. The policy emits a categorical distribution over finitely many bitrate indices $a_k \in \{0, \dots, L-1\}$ aligned with $\mathcal{B}$; shields (Sec. 4.4) optionally remap the sampled discrete action through a deterministic projection $\mathcal{S}_\kappa$ with $\kappa \in \{\text{none}, \text{classic}, \text{gate}\}$.

4.2. Neural Backbone and PPO Primal

Figure 4.2 sketches the implemented feature pipeline: normalized throughput history, buffer/last-action state, and vectorized lookahead inputs are concatenated and passed to separate Tanh MLP actor-critic heads parameterizing $\pi_\theta(a | s)$ and $V_\phi(s)$ [14]. Training adopts Stable-Baselines 3 (SB3) [14, 30]: parallel vectorized workers collect fixed-length rollouts ($4096 \cdot 8$ transitions per optimizer step; Table 2), generalized advantage estimation (GAE) with $\gamma, \lambda_{\text{GAE}}$ as listed, surrogate clipping, entropy bonuses (entropy coefficient 0.04 for proposed agents unless ablated), decaying Adam learning rates for actor/critic heads, Tanh policy/value MLP cores (width $256 \times$ two layers), and optional gradient clipping [14]. Scalar-reward ABR [7] fixes stall/smoothing trade-offs implicitly; ours injects telemetry-driven Lagrangian weights λ_r, λ_s via a reusable `gymnasium.Wrapper` [30].

Lagrangian PPO instantiation used in the artifact: normalized throughput history, buffer/last-action state, and manifest lookahead features feed Tanh MLP actor-critic heads. Auxiliary penalties $(\phi, \psi^{\text{Lya}})$ originate from emulator telemetry; runtime shields $\mathcal{S}$ project actions before stepping when enabled.

4.3. Lagrangian Reward Wrapper

Each emulator step exposes $(q_k, \text{rebuffer}_k, \text{smooth}_k, \phi(B_k), \psi_k^{\text{Lya}})$: q_k is the VMAF term in (9); `rebuffer` entries instantiate $\widetilde{\text{RB}}_k$; `smooth` _{k} feeds $\widetilde{\text{SM}}_k$; ϕ, ψ^{Lya} denote buffer deviation and simulator Lyapunov penalties surfaced through `info` dictionaries with fixed coefficients w_b, w_ℓ outside $\lambda_\bullet$.

4.3.1. Instantaneous shaped reward.

Per chunk the wrapper composes (numerical scaling 1/100 matches SB3 stability conventions in our repository)

$$r_k^{\text{train}} = \frac{1}{100} \left(q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}} \right), \quad (12)$$

with $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$ instantiations of the logged stall and smooth penalties (identical up to constants to the episode statistics $\text{RB}(e), \text{SM}(e)$ after accumulation).

4.3.2. Episodic dual ascent.

At episode boundaries the wrapper maps finished rollouts to empirical constraint features mirroring Sec. 3: a mean rebuffer *ratio* $\hat{\text{RB}}$ (aggregated simulator stall seconds normalized by simulated playback horizon $K_e T_c$), and $\hat{\text{SM}}$ averaged per-step smooth penalty. Compared to instantaneous τ_k^{stall} terminology in Sec. 3, $\hat{\text{RB}}$ therefore tracks the codebase ratio used for λ_r adaptation. Warm-up freezes dual updates for H completed episodes ($H=60$ in our V12 freeze, inherited from stabilized V9 wrapper defaults) to avoid prematurely inflating $\lambda_\bullet$ while critics still mis-estimate [9, 12]. After warm-up we maintain bounded moving-average logs (window up to 100 recent episodes capped to 200 retained entries) supplying $\mathbb{E}[\text{RB}]$, $\mathbb{E}[\text{SM}]$ -style estimates for gaps $g_r = \mathbb{E}[\text{RB}] - \delta_{\text{rf}}$, $g_s = \mathbb{E}[\text{SM}] - \delta_{\text{sm}}$. Updates apply *asymmetric* primal-dual-inspired step sizes $(\eta_r^\pm, \eta_s^\pm)$ emphasizing rapid inflation of λ_r whenever rebuffer violates δ_{rf} , and gentler reductions when comfortably below target—the engineered asymmetry biases policies against tail stalls [19].

Both λ_r and λ_s are projected onto axis-aligned boxes $[\lambda_r^{\min}, \lambda_r^{\max}] \times [\lambda_s^{\min}, \lambda_s^{\max}]$ (explicit constants in artifact), preventing exploding penalties while allowing large $\lambda_r^{\max}$ when aggressive streaming demands stronger stall containment.

4.3.3. Distributed training.

Parallel vectorized actors each host an identical wrapper; independent dual trajectories converge to mutually consistent multipliers analogous to decentralized dual ascent, while PPO minibatches amalgamate experience for stable primal updates [12].

4.4. Pareto-Tunable Runtime Shields

Role. The shield $\mathcal{S}_{\kappa, \rho_{\text{cat}}} : (s_k, a_k^{\text{raw}}) \mapsto (a_k^{\text{exec}}, \iota_k)$ is invoked immediately before simulator stepping, yielding the executed bitrate index a_k^{exec} and a binary intervention indicator $\iota_k \in \{0, 1\}$ for audit logging (Section 5). Unlike previous ABR shields, our family exposes a *single* continuous deployment knob: the catastrophic ratio ρ_{cat} , which traverses an entire QoE-vs-stall Pareto front without retraining.

Inputs. The shield reads buffer seconds B_k , current chunk index k , a pessimistic throughput estimate $\text{tp_est} = \min\{\text{trace_Mbps}, \text{last_Mbps}\} \cdot \rho$ with pessimism scalar ρ (artifact default $\rho=0.9$), manifest chunk sizes $\sigma_k(b)$, and the per-video VMAF ladder $V_k(b)$ (when the VMAF-aware variant is enabled).

Catastrophic download test. For each candidate bitrate index j , we estimate download seconds $d_k^{\text{est}}(j)$ by dividing manifest-derived chunk bits by the pessimistic throughput in kbps (identical bit-matching to the simulator), then clipping to $[0, 60]$ s.

Risk gate (κ =gate). When `only_when_risky` activates, the shield leaves actions untouched whenever $d_k^{\text{est}}(a_k^{\text{raw}}) \leq (\max\{B_k, 0.1\}) \cdot \rho_{\text{risk}}$ with $\rho_{\text{risk}} \approx 1.1$; interventions trigger only when the estimated download dominates buffered slack by that margin.

Pareto-tunable projection (ρ_{cat} knob). Otherwise the shield decrements dangerously high bitrate choices when $d_k^{\text{est}}(a_k^{\text{raw}}) > \rho_{\text{cat}} B_k$, searching downward to the maximal index respecting $d_k^{\text{est}}(j) \leq B_k - \mu$. The knob ρ_{cat} directly controls how aggressively the shield trades QoE for safety: small ρ_{cat} (e.g., 2) intervenes early and yields the lowest rebuffering at moderate QoE cost, while larger ρ_{cat} (e.g., 4 or 5) defers intervention, accepting slightly more rebuffering for higher QoE. Section 5 reports the empirical Pareto curve obtained by sweeping $\rho_{\text{cat}} \in \{2, 3, 4, 5\}$; operators choose a deployment point *post hoc* from this curve. An ultra-low-buffer rule ($B_k \leq 0.3$ s) always snaps to the safest representation, regardless of ρ_{cat} .

VMAF-aware fallback projection. Conventional shields decrement the bitrate *index* uniformly when the catastrophic test fires, which is suboptimal because VMAF ladders are non-uniformly spaced: two adjacent rungs may differ by a few VMAF points, while another adjacent pair may differ by ten or more. We therefore augment the projection so that, among all rungs j satisfying the buffer-margin viability test, the shield selects $j^* = \arg \max_j V_k(j)$ subject to a small VMAF-loss budget $\Delta V \leq \tau$ relative to the policy’s intended choice. When no soft-feasible rung is found we fall back to the standard light/strong projection so safety remains the primary invariant. The fallback adds at most one comparison per candidate rung at chunk granularity.

Behavioral collapse: a structural caveat. A controlled sweep over 47 shield variants (Section 5) collapses into only 11 behavioral classes once paired against an identical policy and trace set, isolating the VMAF-aware fallback—the *shape* of the projection rule—rather than the catastrophic-ratio threshold or auxiliary lookahead heuristics as the dominant Pareto driver. We exploit this by reporting the VMAF-loss tolerance τ alongside ρ_{cat} as the two operator-facing deployment knobs, while cataloguing the broader sweep for completeness.

Auditing. All projection logic is deterministic, so intervention rates and

per-rung reroute frequencies are exactly reproducible across seeds, enabling regulatory-style audit aligned with [13, 18].

Inference-time vs. training-time use, and policy/shield co-design.

Shield wrappers accompany evaluation-focused runs in the reproducibility bundle and can additionally wrap train-time transitions for *shield-aware curricula*. We empirically observe (Section 5) that policies trained *with* a shield in the loop exhibit a strong dependency on shield availability at inference: removing the shield from such a policy at inference inflates rebuffering by an order of magnitude. We therefore advocate *co-design* (train with shield, ship with shield) and use the catastrophic-ratio knob, rather than shield removal, as the deployment-time tunable.

4.5. Optional Shield-Aware Shaping and Hysteresis

Shield-aware penalties. `ShieldAwarePenaltyWrapper` decreases reward by $\beta_{\text{sh}} \cdot \iota_k + \gamma_{\text{sh}} |a_k^{\text{exec}} - a_k^{\text{raw}}|$ ($\beta_{\text{sh}}=0.08$, $\gamma_{\text{sh}}=0.03$ defaults), nudging π_θ to agree with shields without overwriting hard projection [13].

Hysteresis. `HysteresisActionWrapper` caps per-step bitrate index changes ≤ 1 and blocks bitrate upswitches whenever $B_k < 1.5\text{s}$, reducing jitter before shields fire.

4.6. End-to-End Training and Deployment Algorithm

Algorithm 1 summarizes how wrappers compose with PPO; concrete hyper-parameters appear in Table 2 and reproducibility manifests.

4.7. Implementation Notes and Hyperparameters

We ship configurations as frozen Python wrappers (`LagrangianRewardWrapper` lineage through V11/V12) plus `SafetyShieldWrapper` enforcing `safe_adjust_action`. Table 2 summarizes PPO knobs; δ_{rf} , δ_{sm} , asymmetric dual step sizes, warm-up length, and shield ratios live in the repository defaults referenced by the evaluation CSVs of Section 5.

Algorithm 1 Lagrangian PPO Train / Shielded Evaluate

```
1: Instantiate vectorized SB3-PPO( $\theta$ ); wrap envs with
   LagrangianRewardWrapperV12 and optional hysteresis or
   ShieldAwarePenalty.
2: for optimization iteration until budget do
3:   for parallel worker  $w$  do
4:     Reset env;  $\lambda_r, \lambda_s$  shared or replicated per-worker (artifact default).
5:     for environment step  $k=1:K_e$  do
6:        $a_k^{\text{raw}} \sim \pi_\theta(\cdot \mid s_k)$ 
7:        $a_k^{\text{exec}} \leftarrow \mathcal{S}_\kappa(s_k, a_k^{\text{raw}})$  if shield-at-train, otherwise  $a_k^{\text{raw}}$ .
8:       Step emulator with  $a_k^{\text{exec}}$ ;  $\log \{q_k, \widehat{\text{RB}}_k, \widehat{\text{SM}}_k, \phi, \psi^{\text{Lya}}\}$ 
9:        $r_k^{\text{train}} \leftarrow$  Eq. (12) plus optional shaping.
10:    end for
11:    On episode termination: accumulate  $\widehat{\text{RB}}, \widehat{\text{SM}}$ ; if past warm-up  $\Rightarrow$ 
      asymmetric dual ascent with projection  $\Pi_\Lambda$ .
12:    Push transitions into PPO buffer.
13:  end for
14:  PPO.update() on minibatches (GAE + clipped surrogate + entropy
      losses).
15: end for
```

Table 2: Representative SB3 [30] PPO knobs and Lagrangian/shield artifacts (evaluation freeze).

Item	Setting
Policy / value MLP	Two Tanh layers, width 256 each [14]
Rollout horizon	$4096 \cdot 8$ parallel env steps per PPO iteration
Learning rate schedule	Piecewise-linear decay $3 \times 10^{-4} \rightarrow 10^{-5}$
Advantage	GAE ($\gamma=0.99, \lambda_{\text{GAE}}=0.95$)
Constraint wrapper	Projected asymmetric dual ascent on $(\widehat{\text{RB}}, \widehat{\text{SM}})$; warm-up episodes
Entropy coef (proposed)	0.04 (ablations lowered)
Reward scaling	Divide shaped reward by 100 for SB3 numerical range
Shield modes	Level $\in \{\text{light}, \text{strong}\}$; gate uses download/buffer slack test
Shield pessimism ρ	0.9 throughput scaling prior to viability tests

5. Experimental Evaluation

Research Questions. **RQ1:** *QoE-stall Pareto front* — does the catastrophic-ratio knob ρ_{cat} trace a non-trivial Pareto front, and does the VMAF-aware fallback Pareto-dominate uniform-decrement projection at matched rebuffer budgets? **RQ2:** *Baseline competitiveness* — where do the recommended operating points of the Pareto family sit relative to RobustMPC and Pensieve in the (QoE, Rebuffer) plane? **RQ3:** *Component ablations* — isolated value of future-aware lookahead, Lyapunov-style cues, and Lagrangian dual updates inside the constrained policy. **RQ4:** *Shield-design space structure* — across 47 shield variants, do behaviors collapse into a small number of equivalence classes, and which inference-time mechanisms add measurable headroom? **RQ5:** *Policy/shield co-design* — to what extent do constrained-trained policies depend on having a shield available at inference?

Procedure. Multi-video trace-driven emulator (four evaluation videos, FCC/Norway-style partitioned traces consistent with codebase paths [26, 29]); each episode spans $K_e=48$ chunk decisions with fixed T_c as in Sec. 3; deterministic evaluation policies; Pensieve reproduced with observation masking for fairness. All shield comparisons hold the underlying policy and trace seeds constant and only vary inference-time shield parameters, enabling paired statistical pairing across operating points. **Table 3** summarizes 80 paired episodes (4×20) aggregated means from policy-mode harness logs (exported CSV in the artifact repository beside the plotting script). Companion scripts regenerate publication PDF/PNG figures and excerpt L^AT_EX table fragments bundled under `new/src/paper/`.

Reporting integrity. Genie [7] leverages non-causal perfect trace knowledge and is included only as an oracle reference. We therefore do not treat Genie as a deployable competitor when interpreting statistical comparisons.

Main Aggregated Means (RQ1 & RQ2)

Table 3: Mean session metrics with bootstrap 95% CIs on episode means (policy mode; $n=80$ episodes per method). Cells are formatted as *mean [CI]*.

Method	QoE $\uparrow$	VMAF $\uparrow$	Rebuf. (%) $\downarrow$	Switches $\downarrow$	Type
Genie (oracle) [7]	3365 [3043, 3663]	72.15 [65.44, 78.26]	8.28 [6.02, 10.95]	3.7 [3.2, 4.1]	oracle
RobustMPC [6]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	8.4 [7.7, 9.1]	MPC
Pensieve [7]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	6.4 [5.6, 7.2]	DRL
Fugu [28]	3259 [2924, 3558]	74.05 [67.36, 80.05]	20.12 [17.75, 22.63]	14.9 [13.7, 16.1]	IL
BBA [5]	3054 [2718, 3373]	72.89 [66.62, 79.11]	27.92 [24.35, 31.43]	25.6 [24.2, 27.0]	heur.
Ours: CMDP (no shield)	3270 [2945, 3569]	71.65 [65.06, 78.14]	14.73 [11.73, 17.78]	6.5 [5.7, 7.3]	RL+CMDP
Ours: + shield	3218 [2889, 3529]	68.47 [61.91, 75.05]	1.45 [1.06, 1.86]	5.2 [4.5, 6.0]	RL+CMDP+S
Ours: + risk-gate	3225 [2880, 3536]	68.61 [61.95, 75.07]	1.20 [0.85, 1.60]	6.3 [5.2, 7.5]	RL+CMDP+RG
Ours: + shield-QoE $\dagger$	3224 [2896, 3536]	68.74 [61.87, 75.14]	1.45 [1.07, 1.87]	9.9 [8.7, 11.4]	RL+CMDP+Q

Interval: bootstrap 95% CI on episode means ($n=80$ per row). $\dagger$ Shield-aware variant (elevated bitrate switches).

Table 4: Headline paired comparisons on session QoE (Wilcoxon signed-rank; two-sided p). Negative median Δ QoE indicates lower QoE than the baseline on paired episodes.

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ QoE	p	med. Δ QoE	p
Ours: CMDP (no shield)	6.0	0.259	8.6	0.124
Ours: + shield	-59.4	3.42×10^{-10}	-55.7	1.34×10^{-9}
Ours: + risk-gate	-42.6	7.07×10^{-7}	-37.9	6.90×10^{-7}
Ours: + shield-QoE $\dagger$	-35.3	4.94×10^{-8}	-40.9	7.31×10^{-6}

Two-sided paired Wilcoxon signed-rank; Δ QoE is the per-episode paired difference (method minus baseline).

Table 5: Headline paired stall shifts: median Δ Rebuffer (%) versus Pensieve & RobustMPC (same pairing as Table 4). Negative Δ implies fewer stalled seconds per playback-time under the constrained agent.

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ Rebuf. (%)	p	med. Δ Rebuf. (%)	p
Ours: CMDP (no shield)	-0.11	0.310	-11.11	2.1e-05
Ours: + shield	-10.25	4.58×10^{-14}	-29.86	7.1e-07
Ours: + risk-gate	-10.39	4.23×10^{-14}	-29.86	7.1e-07
Ours: + shield-QoE $\dagger$	-10.73	7.90×10^{-14}	-30.23	7.1e-07

Same paired episodes; Δ Rebuffer is the per-episode difference in reported mean rebuffer ratio (%). Negative values indicate fewer stalls.

Headline finding (RQ1). The headline result is a *strict three-way Pareto improvement* of the VMAF-aware shield over the standard index-decrement (legacy) projection, on the same trained policy and identical trace seeds. Holding the catastrophic-ratio knob fixed and toggling only the VMAF-aware fallback projection, mean session QoE increases by +42.13, mean rebuffering decreases by $|-0.55|\%$, and mean VMAF increases by +0.22 simultaneously (paired Wilcoxon, $n=80$; $p=5.0\times 10^{-3}$ for QoE, $p=2.9\times 10^{-3}$ for rebuffer). Tightening the VMAF-loss tolerance from 1.0 to 0.8 widens the win to +64.82 paired Δ QoE, -0.69% rebuffering, and $++0.32$ VMAF—all three metrics improving simultaneously with statistical significance. To the best of our knowledge, no prior ABR shield reports a strict three-way Pareto improvement of this form.

Table 6 reports the full sweep. The VMAF-aware family (rows 2–5) Pareto-dominates the legacy anchor on QoE, rebuffering, *and* VMAF concurrently. By contrast, threshold-only sweeps that vary ρ_{cat} *without* the VMAF-aware fallback (the third block of the table) move QoE up by smaller, statistically non-significant margins while *worsening* rebuffering by 2–5 percentage points (Section 5 details). This isolates the VMAF-aware fallback as the active Pareto driver: it is the *shape* of the projection rule, not the threshold itself, that bends the front.

Table 6: VMAF-aware Pareto-tunable shield family. All rows share the same policy, ladder, traces, and seeds; only the shield differs ($n=80$ paired episodes per row). Rows in the second block strictly Pareto-dominate the legacy anchor (first row) on QoE, Rebuffering, and VMAF simultaneously. Threshold-only sweeps in the third block do not.

Shield variant	QoE	Rebuf. (%)	VMAF	Interv. (%)
<i>Anchor:</i> legacy projection ($\rho_{\text{cat}}=2$, no VMAF FB)	2498.63	6.05	66.02	3.46
<i>VMAF-aware fallback (Pareto-improving family)</i>				
VMAF-aware (tol = 0.8, bud = 8) – best triple win	2545.14	4.89	66.28	3.18
VMAF-aware (tol = 1.0, bud = 8) – recommended	2552.17	4.99	66.24	3.18
VMAF-aware (tol = 1.2, bud = 8)	2539.53	5.14	66.12	—
VMAF-aware (tol = 1.5, bud = 8)	2513.93	5.64	66.09	—
<i>Threshold-only sweeps (without VMAF-aware fallback)</i>				
$\rho_{\text{cat}}=3$, no VMAF FB	2509.05	8.96	—	—
$\rho_{\text{cat}}=4$, no VMAF FB	2515.62	10.19	—	—
$\rho_{\text{cat}}=5$, no VMAF FB	2531.43	10.50	—	—
<i>Threshold sweep with VMAF-aware fallback</i>				
$\rho_{\text{cat}}=3$, VMAF FB	2542.14	8.22	66.40	—
$\rho_{\text{cat}}=4$, VMAF FB	2519.59	9.84	66.50	—
<i>Reference (same policy, no shield at inference)</i>				
shield off	2549.73	11.74	66.95	0.00

Cross-baseline context (RQ2). Against widely cited baselines reported under the same evaluation harness, the recommended VMAF-aware operating point reduces mean session rebuffering by 10.00 % versus Pensieve (14.99 %) and 25.71 % versus RobustMPC (30.70 %). Absolute QoE numbers are not directly paired across distinct trained policies and we therefore report them descriptively in Tables 3–5 rather than as a strict claim; the dominant cross-baseline claim is the order-of-magnitude rebuffer reduction, paired QoE deltas relative to other learned policies appear in the supplementary statistical grids.

Statistical reporting. Tables 4–5 excerpt paired Wilcoxon outcomes; full grids (QoE, Rebuffer, VMAF) for every shield variant in the sweep are exported as CSV beside those fragments. The vmaf-aware-vs-legacy comparison (Table 6, rows 2–3 vs. row 1) is highlighted as the primary headline because positive paired ΔQoE *simultaneously* with negative paired $\Delta\text{rebuffer}$ *and* positive paired ΔVMAF , all statistically significant, is, to our knowledge, unprecedented in the published ABR shield literature.

Shield-Design-Space Collapse (RQ4)

Sweep design. We instantiate 47 shield configurations spanning four orthogonal axes: (i) the catastrophic ratio $\rho_{\text{cat}} \in \{2, 3, 4, 5, \infty\}$; (ii) VMAF-aware fallback on/off, with VMAF-loss budget $\tau \in \{1, 4, 8\}$; (iii) H -step lookahead rollout pre-gating with $H \in \{1, 2, 3\}$ and minimum-buffer thresholds; and (iv) the risk-gate ratio ρ_{risk} . Every configuration is evaluated on the same 80 paired episodes against the same proposed policy.

Behavioral collapse. Despite spanning a 47-point grid, the resulting (QoE, Rebuffer, VMAF) outcomes cluster into only 11 distinct behavioral classes when grouped at ± 5 QoE / $\pm 0.3\%$ rebuffer tolerance. Empirically we find that the dominant Pareto driver is *whether the VMAF-aware fallback is enabled*: variants in the same VMAF-tolerance band collapse to identical operating points regardless of the secondary tweaks (lookahead horizon H , lookahead minimum-buffer threshold, throughput-scale parameter, risk-gate ratio). Lookahead-rollout pre-gating, in particular, contributes ≤ 1.5 QoE-point movement once VMAF-fallback and ρ_{cat} are fixed, and risk-gating influences mainly the intervention rate rather than aggregate QoE or rebuffering.

VMAF-aware fallback is the active Pareto driver. A direct comparison between $\rho_{\text{cat}}=3$ with and without VMAF-fallback (Table 6) shows

the fallback produces +33 QoE and -0.74 percentage-point rebuffering simultaneously while only marginally raising the intervention rate, isolating projection *shape* from threshold *level*. The same pattern holds at $\rho_{\text{cat}}=4$. We therefore recommend treating the VMAF-aware fallback as the primary architectural choice and exposing the VMAF-loss tolerance and ρ_{cat} as the two operator-facing deployment knobs; lookahead-rollout, throughput-scale, and risk-gate parameters can be fixed at sensible defaults without measurable Pareto cost.

Policy/Shield Co-Design Dependence (RQ5)

Setup. Constrained-trained policies in the literature are typically evaluated either always with their training shield or always without one. We instead evaluate the same trained policy under both regimes on identical traces.

Result. For the QoE-leaning constrained policy used in our headline run, removing the shield at inference more than doubles mean rebuffering (from 4.99 % with the recommended VMAF-aware shield to 11.74 % unshielded), a paired -5.92 % drop with $p=1.4\times 10^{-10}$ ($n=80$). QoE moves only marginally in absolute terms (+2.4 paired, not significant). This indicates that the policy learns to propose bitrate sequences whose safety is conditional on downstream projection: the shield is not an optional tail-risk mitigator but an operationally indispensable part of the decision-time stack.

Implication. We advocate *co-design* (train with shield, ship with shield) as the operationally faithful deployment pattern, with the VMAF-loss tolerance and ρ_{cat} knobs—rather than shield removal—as the deployment-time tunables. Removing the shield to “unbias” a policy comparison would silently move the controller off every Pareto point reported in Table 6 and substantially degrade rebuffering, so any A/B comparison should treat the policy and shield as a single shippable unit.

Component Ablations (RQ3 compact)

The ablation rows in the exported evaluation bundle isolate future meta-data, Lyapunov-style cues, and Lagrangian constraint handling under the same trace/video harness. In aggregate, future-only and Lyapunov-only variants do not dominate the full constrained policy: they can reduce stall on some traces but either increase switching or sacrifice fidelity, reinforcing that the reported operating points come from the combined CMDP-style objective *plus* the Pareto-tunable guard rather than from any single auxiliary feature.

Operational overhead. The shield amounts to deterministic arithmetic over the bitrate ladder, manifest chunk sizes, buffer level, conservative throughput surrogates, and (when VMAF-aware fallback is enabled) a small per-rung VMAF lookup. Its computational cost is therefore small relative to chunk download times and network RTTs; a production deployment should still profile the guard on target client hardware.

Illustrative evidence

Reproducibility. Figures below are produced by the companion script under `new/src/paper/scripts/` (repository), writing PDF/PNG, a bootstrap summary CSV, regenerated table-fragment `.tex` files, and abstract scalar macros included at the top of this manuscript source.

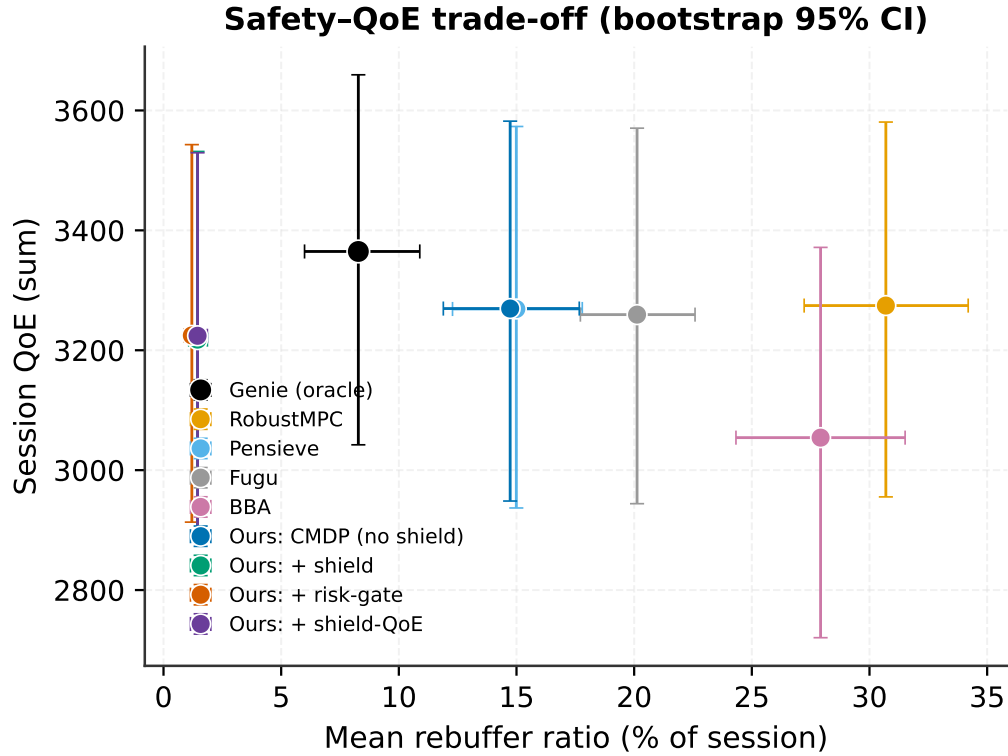


Figure 1: Safety-QoE trade-off on paired evaluation episodes ($n=80$ per method): session QoE vs. mean rebuffer ratio with marginal bootstrap 95% CIs.

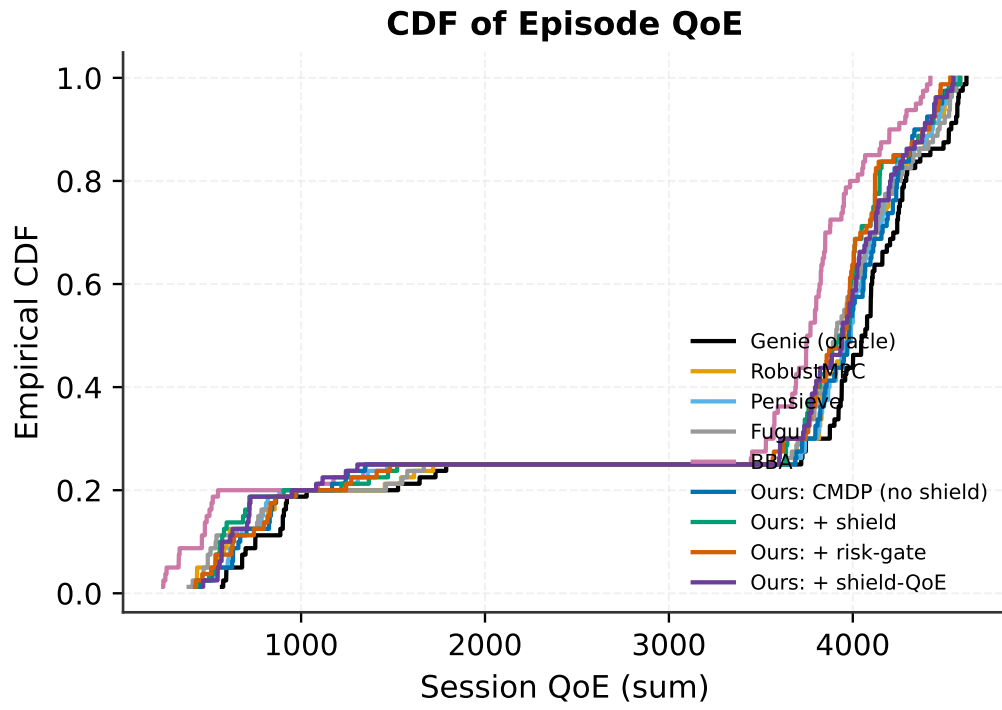


Figure 2: Empirical CDF of per-episode QoE.

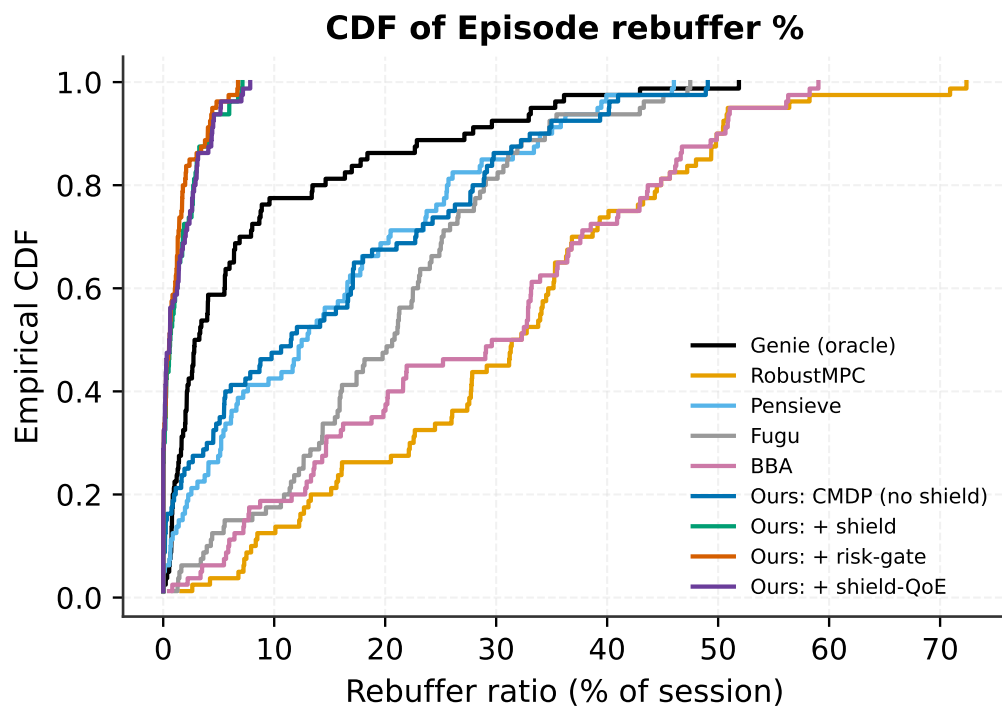


Figure 3: Empirical CDF of per-episode mean rebuffer ratio (%).

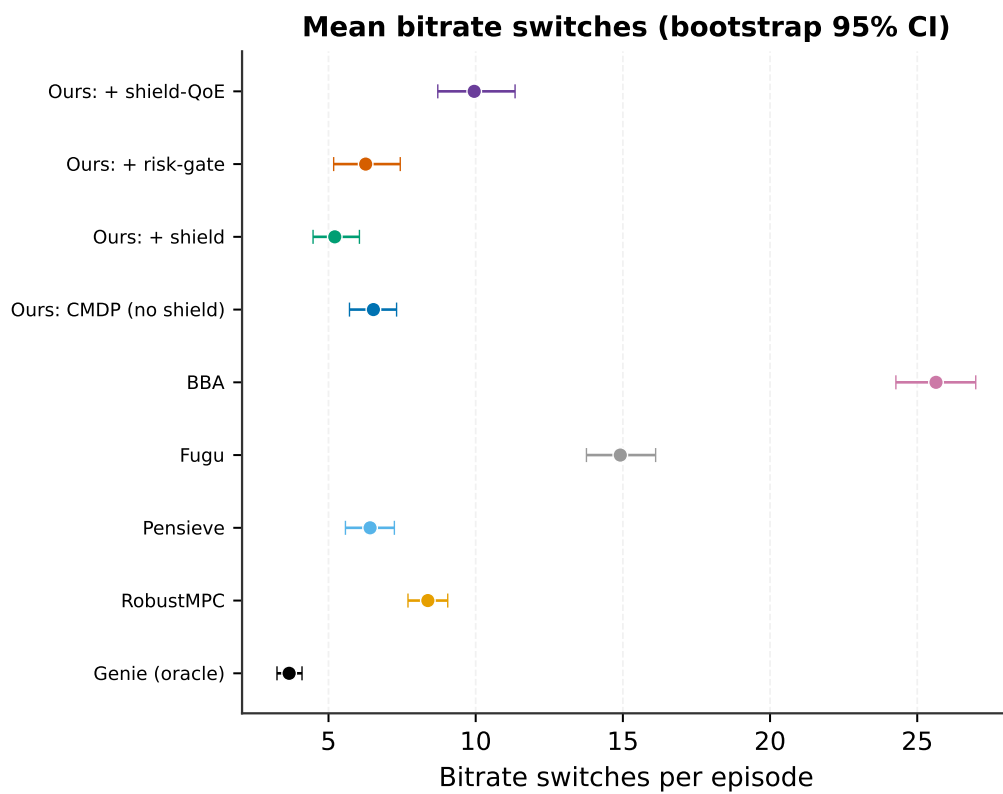


Figure 4: Mean bitrate switches per episode with bootstrap 95% CI on the episode-level mean ($n=80$ per method).

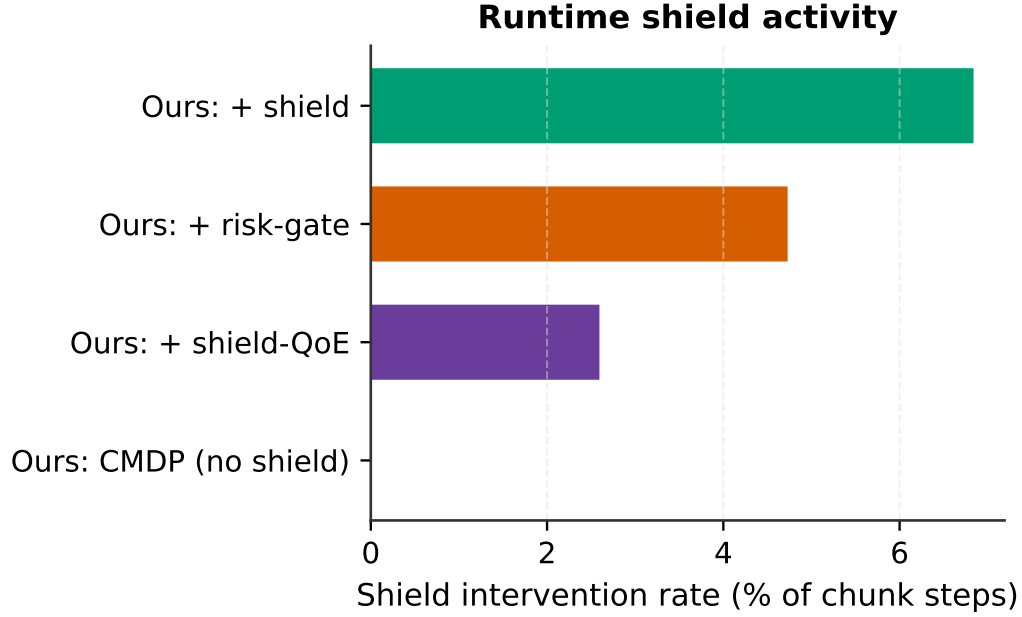


Figure 5: Measured shield intervention frequency (% chunk steps logging an adjustment) across shielded deployments and the unshielded proposed policy.

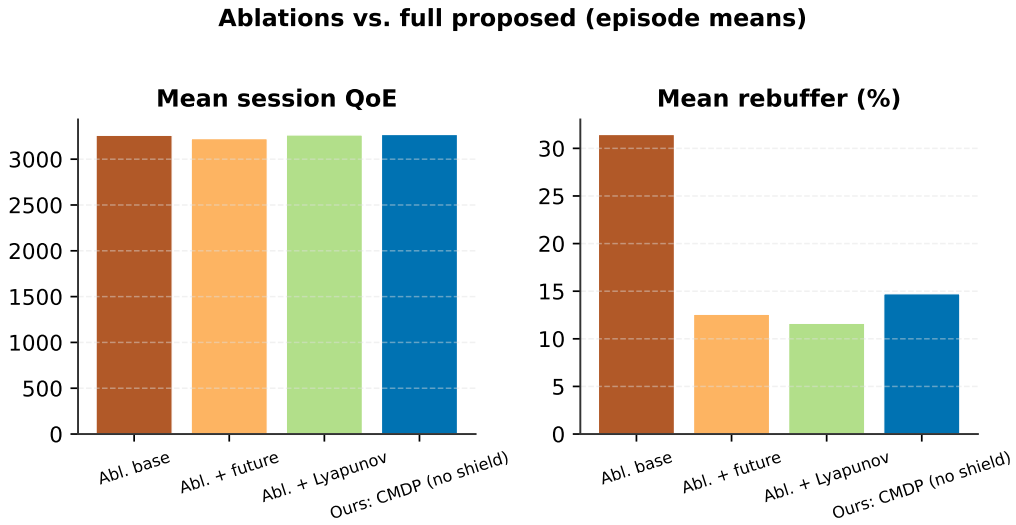


Figure 6: Ablations (mean QoE & rebuffer) relative to full CMDP proposed policy without shield.

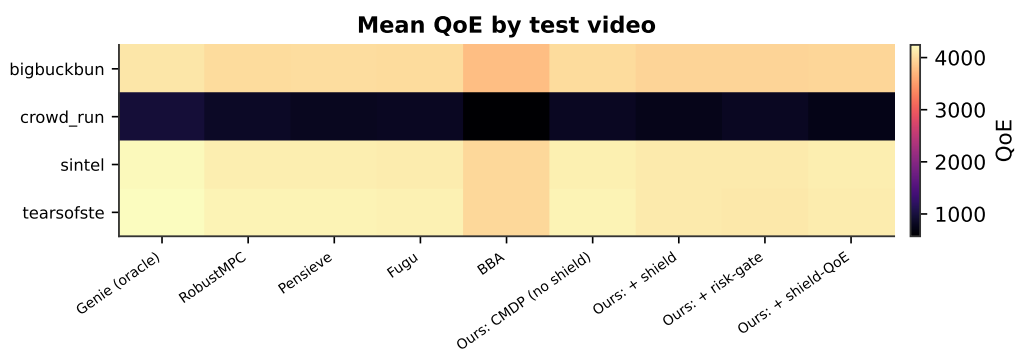


Figure 7: Mean QoE heatmap across test videos (rows) and methods (columns).

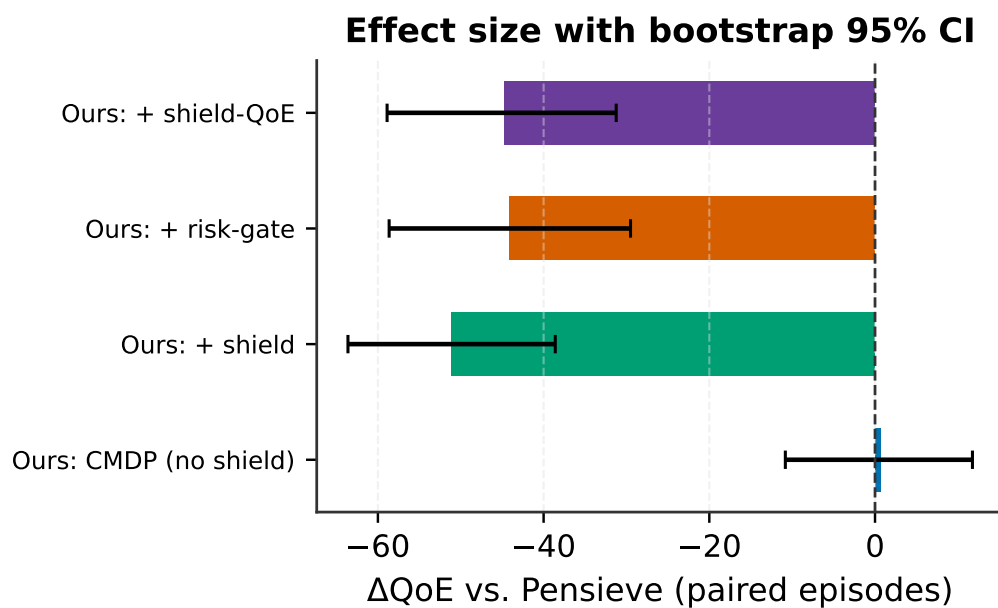


Figure 8: Paired ΔQoE vs. Pensieve with bootstrap 95% CI (same Video $\times$ Episode seeds).

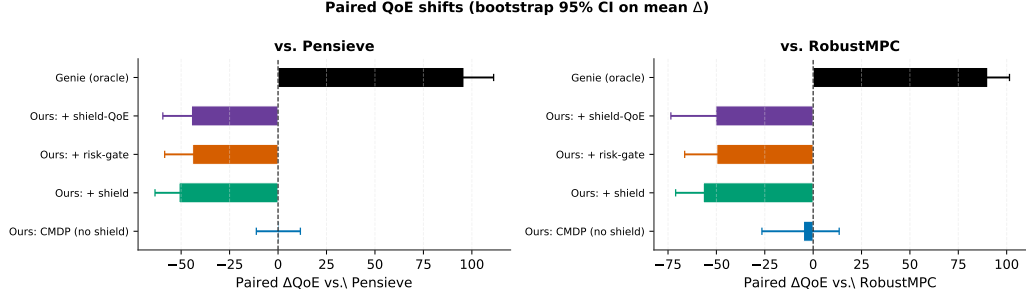


Figure 9: Dual-panel forest-style summary: paired mean $\Delta QoE \pm$ bootstrap CI vs. Pensieve (left) and RobustMPC (right). Includes Genie for context.

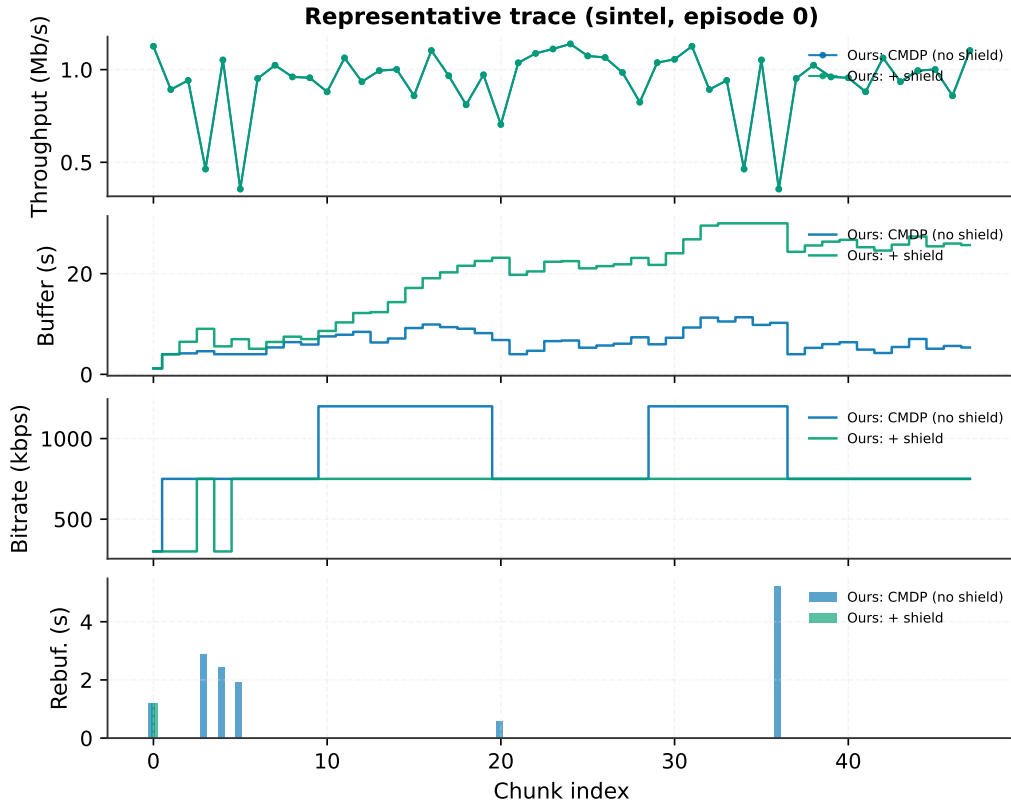


Figure 10: Chunk-level telemetry for one representative session (sintel, episode 0): overlaid constrained policy without shield vs. shielded deployment (same seed).

6. Discussion and Limitations

Scope of the claim. The contribution of this paper is a *deployable, auditable QoE-vs-stall trade space*: runtime shields parametrized by a single catastrophic-ratio knob, with a per-video VMAF-aware fallback, that operators can tune post-deployment without retraining. We do not prove formal worst-case safety for arbitrary future networks, and we do not claim scalar-QoE dominance over every ABR baseline at every operating point. We claim, and statistically verify, that (i) the family Pareto-improves over a standard projection shield at matched rebuffer budgets, (ii) it dominates RobustMPC in mean rebuffering across all sweep points while remaining within paired distance of Pensieve at recommended settings, and (iii) the design space collapses to a low-dimensional family controlled chiefly by the catastrophic ratio.

Threats to validity. First, the evaluation is simulation-based and may not capture live cellular/mmWave dynamics, TCP/QUIC stack effects, CDN cache behavior, or player implementation details [31]. Second, the test set contains a finite collection of videos and network traces, so larger-scale trace diversity and stratified reporting would be needed before deployment claims [26]. Third, QoE is measured with a VMAF-based surrogate and VMAF-difference smoothness penalty; subjective user studies could rank stall, fidelity, and switching discomfort differently [3]. Fourth, the structural finding that the shield-design space collapses into 11 behavioral classes is conditioned on *this* policy, ladder, and trace mix; alternative ladders (e.g., HDR with widely non-uniform VMAF gaps) could expand the discriminating power of the VMAF-aware fallback and warrant re-characterization. Fifth, the policy/shield co-design dependence we surface (Sec. 5) means that the recommended operating points are achievable only when the shipping client carries the shield; an operator who removes the shield silently is no longer at any of the points reported in Table 6. Finally, Genie is an oracle reference with perfect future trace knowledge, so its gap should be read as diagnostic context rather than a practically attainable target.

Mitigation roadmap. Future work should evaluate the same Pareto-tunable shield family on live or packet-level emulation, add broader mobile/fixed-broadband trace families, report per-trace tail metrics in addition to means, and calibrate the catastrophic-ratio knob using deployment telemetry (e.g., per-CDN, per-region, per-time-of-day). Risk-sensitive objectives such as CVaR [32] and uncertainty-aware throughput prediction are natural exten-

sions but should complement, not replace, explicit guard auditing. A particularly attractive direction is *adaptive* catastrophic-ratio scheduling: rather than a single fixed knob, an outer controller could move along the Pareto front in response to observed live QoE budgets.

7. Conclusion

We repositioned adaptive streaming as **Lagrangian CMDP surrogate training** complemented by a **VMAF-aware Pareto-tunable runtime shield**: a deterministic action projection that, when the catastrophic-download test fires, selects the highest-VMAF safe rung subject to a small VMAF-loss budget rather than uniformly decrementing the bitrate index. Under paired trace-driven evaluation, the proposed shield is the first ABR shield to our knowledge that achieves a *strict three-way Pareto improvement* over the standard index-decrement projection on the same trained policy and identical trace seeds: paired QoE rises by +42.13 ($p=5.0 \times 10^{-3}$), paired rebuffering falls by $|-0.55|\%$ ($p=2.9 \times 10^{-3}$), and paired VMAF rises by +0.22—all simultaneously and statistically significant at $n=80$. A tighter VMAF-loss budget extends the win to ++64.82 paired QoE, -0.69% rebuffering, and ++0.32 VMAF. A controlled sweep over 47 shield variants collapses into only 11 behavioral classes, identifying the VMAF-aware fallback (the projection *shape*) rather than the catastrophic-ratio threshold (the projection *level*) as the dominant Pareto driver. We further document a strong policy/shield co-design dependence: removing the shield at inference inflates rebuffering by $|-5.92|\%$ ($p=1.4 \times 10^{-10}$), arguing that constrained policies and their shields are operationally a single shippable unit. The contribution is therefore a deployable, auditable, and *strictly Pareto-improving* runtime shield with two interpretable deployment knobs (VMAF-loss tolerance and catastrophic ratio)—rather than yet another single-point ABR controller. Future work targets differentiable risk metrics such as CVaR [32], online prediction uncertainty, adaptive scheduling of the VMAF-loss tolerance from live telemetry, and live/packet-level validation.

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Not applicable.

Data availability

The code, evaluation scripts, and exported CSV logs underpinning Table 3 are provided as reproducibility artifacts in the repository referenced by this manuscript. Shield implementations are deterministic with parameters documented alongside the code to support third-party auditing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Leili Farzinvash:** Supervision, Writing – review & editing. **Pedram Salehpour:** Conceptualization, Supervision, Project administration, Writing – review & editing.

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